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Topic: Comparison of Deep Learning Architectures

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Comparison of Deep Learning Architectures

Feedforward Neural Networks, Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs)

Description

This project studies three fundamental deep learning architectures — Feedforward Neural Networks, Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs) — and examines the type of problems each architecture is best suited to solve. Understanding the structural differences between these architectures helps in selecting the right model for a given task, whether it involves simple tabular data, image data, or sequential data.

Task 1: Basic Architecture of Feedforward Networks, CNNs, and RNNs

1.1 Feedforward Neural Network (FNN)

A Feedforward Neural Network is the simplest type of artificial neural network. Data moves in only one direction — from the input layer, through one or more hidden layers, to the output layer — with no cycles or loops.

- Composed of an input layer, one or more hidden layers, and an output layer.
- Each neuron in a layer is fully connected to every neuron in the next layer (dense/fully connected layers).
- Uses activation functions (ReLU, sigmoid, tanh) to introduce non-linearity.
- Has no memory of previous inputs — each input is processed independently.

1.2 Convolutional Neural Network (CNN)

A CNN is designed to process grid-structured data, most commonly images, by exploiting spatial locality through convolution operations.

- Convolutional layers apply learnable filters/kernels that slide over the input to detect local features (edges, textures, shapes).
- Pooling layers (max pooling/average pooling) reduce the spatial dimensions and computational load while retaining important features.
- Fully connected layers near the end combine extracted features to produce the final classification or output.
- Parameter sharing (the same filter is reused across the image) makes CNNs efficient for high-dimensional image data.

1.3 Recurrent Neural Network (RNN)

An RNN is designed to handle sequential data by maintaining a hidden state that carries information from one time step to the next.

- Neurons have recurrent (looping) connections, allowing outputs from previous steps to be fed back as inputs.
- A hidden state acts as a form of memory, capturing context from earlier elements in the sequence.
- The same weights are shared across all time steps (parameter sharing across time).
- Variants like LSTM and GRU address the vanishing gradient problem in long sequences.

Task 2: Suitable Applications for Each Architecture

Feedforward Neural Networks

- Structured/tabular data classification and regression (e.g., predicting house prices).
- Simple pattern recognition tasks.
- Credit scoring and risk assessment.
- Used as fully connected layers inside larger CNN or RNN architectures.

Convolutional Neural Networks (CNNs)

- Image classification (e.g., identifying objects in photos).
- Object detection and facial recognition.
- Medical image analysis (X-rays, MRI scans).
- Video processing and frame-level analysis.

Recurrent Neural Networks (RNNs)

- Natural language processing tasks such as language modelling and text generation.
- Machine translation.
- Speech recognition.
- Time-series forecasting (e.g., stock prices, weather data).

Comparison of Architectures

The table below compares the three architectures based on structure, input data, key characteristics, and applications.

Aspect	Feedforward Neural Network	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Structure	Neurons arranged in fully connected layers; signals flow in one direction only, from input to output, with no loops or memory.	Uses convolutional layers with learnable filters/kernels, followed by pooling layers to reduce dimensionality, and fully connected layers near the output.	Neurons have connections that loop back on themselves, allowing information from previous time steps to influence the current output (hidden state/memory).
Input Data	Fixed-size vectors of independent features (e.g., tabular data, flattened numeric values).	Grid-like data such as images (2D pixel grids) or video frames, where spatial relationships between neighbouring values matter.	Sequential data such as text, time-series, or audio, where the order of data points matters.
Key Characteristic	Simple, no parameter sharing, no memory of previous inputs.	Parameter sharing through filters; captures local spatial patterns like edges, shapes, and textures.	Parameter sharing across time steps; captures temporal/sequential patterns and dependencies.
Applications	Basic classification/regression	Image classification, object detection, facial recognition,	Language modelling, machine translation, speech

	tasks, simple pattern recognition, credit scoring, and as building blocks inside larger networks.	medical image analysis, video processing.	recognition, sentiment analysis, time-series forecasting.
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Conclusion

Feedforward Neural Networks are best suited for simple, structured data without spatial or sequential relationships. CNNs excel at processing grid-like data such as images by exploiting spatial locality through convolution and pooling. RNNs are designed for sequential data, using recurrent connections and hidden states to capture temporal dependencies. Choosing the right architecture depends on the nature of the input data and the specific problem being solved.